End-to-End Experimental Validation Workflow: AST-Route Testbed (97 Queries, 4 Scales, 6 Engines)

PHASE 1: Multi-Scale TPC-H Data Provisioning (SF = 1, 10, 30, 100 GB)

- 4 Standard Scales: SF1 (8.7M rows), SF10 (86.6M), SF30 (259.8M), SF100 (866.1M)
- Local Columnar: ZSTD Parquet + DuckDB Native + Apache Arrow Buffers
- Local Row-Store Baseline: SQLite 3 Full B-Tree with PK/FK/Date Indexes
- Cloud Serverless: Google Cloud Storage -> BigQuery Unopt vs. Clustered

PHASE 2: 75-Query Factorial Suite (L1-L5) + 22 Official TPC-H (Q1-Q22)

- Suite A (75 Queries): 25 Canonical Seeds (L1-L5) x 3 Horizons (V1, V2, V3) + Strict Unseen Holdout Split (20 Dev Seeds S1-S4 vs. 5 Locked Unseen S5)
- Suite B (22 Queries): Official Standard TPC-H Benchmark Queries (Q1-Q22)
- Total Evaluated Workload: 97 Distinct SQL Queries (100% Real Execution)

PHASE 3: Static AST Profiling, 10,000-Trial CSCI Sensitivity Audit & 4-Pillar GPU/CPU Fairness Protocol

- Parses all 97 SQL queries via SQLGlot into 9D vectors $x_q = [D, N, J, S, W, U, C, CSCI, \sigma(q)]^T$ (Factorial Mean CSCI = 29.66; Official TPC-H Mean = 26.52)
- 10,000-Trial Monte Carlo CSCI Perturbation Stability: Spearman ρ = 0.9963 (+-10%), 0.9880 (+-20%), 0.9555 (+-50%), Equal-Weight ρ = 0.9812
- 4-Pillar Fairness & Sensor Polling (200-250 Hz psutil + NVML + cudaEvent): Strict 64-bit FP64 Parity (< 1e-9), Unrestricted Multi-Core SIMD, Spill & Energy (J)

Paradigm 1: Row-Store

- Engine: SQLite 3 (ANSI C)
- Storage: Full-Scale B-Tree
- Tuple-at-a-Time Volcano Model
- SF1 Mean: 8417.33 ms
- 95% CI: [6412.2, 10819.7] ms
- Mean Energy: 126.26 J (4.05 MB/J)

Paradigm 2: Serverless Cloud

- Engine: Google BigQuery
- Capacitor Columnar Storage
- Cache Strictly Disabled
- SF1 Clustered: 745.39 ms
- SF100 Clustered: 2571.63 ms
- V1 Pruning: 78.4% (1.47 – 147/1M)

Paradigm 3: Vectorized CPU

- 1) DuckDB: 183.74 ms (SF1) TPC-H Q1-Q22 Mean: 22.82 ms
- 2) Polars: 82.47 ms (SF1, 3.17 J) 95% CI: [67.9, 98.7] ms
- 3) DataFusion: 96.30 ms (3.39 J)
- Out-of-Core Spill: 0->3302 MB

Paradigm 4: SIMT CUDA GPU

- Engine: PyTorch CUDA Tensor
- GPU: NVIDIA RTX 2050 (4GB)
- SF1 Cold: 56.96 ms (0.65 J)
- SF1 Warm: 22.26 ms (0.25 J)
- Multi-Scale PCIe Streaming: 1x (SF1), 4x (SF10), 12x, 40x (SF100)

PHASE 5: Seed-Clustered 95% CIs, LOSO-CV & Unseen Holdout

- Exact FP64 Equivalence EX = 1 (97/97) | Friedman $\chi^2_F(5)$ = 321.50
- 25-Seed LOSO-CV: 54.92 ms (0.00% Median Regret) | Unseen S5: 73.33% Acc
- Official TPC-H Q1-Q22: 72.73% Top-1 Acc (90.91% Near-Opt, 15.84 ms)

PHASE 6: Environment Manifest & Cryptographic SHA-256 Audit

- Automated Booktabs Tables (Tables 1-13) & Unified 300 DPI Vector Figures
- Full Hardware/Software Environment Manifest & Zero-Discrepancy Audit
- Open-Source Repository: github.com/preecha-msu/bigquery-olap-benchmark